[Exp] Multi-Label ICD-10 Classification of Greek Cardiology Discharge Summaries

System Description for the ElCardioCC 2026 Shared Task

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Abstract. We present our system for the ElCardioCC 2026 shared task on automatic ICD-10 coding of Greek cardiology discharge summaries. The system combines six components: a fine-tuned Greek BERT model, two XLM-RoBERTa variants (base and large), a hybrid information retrieval module, a dictionary-based rule system, and a named entity recognition with entity linking pipeline. These are fused under a metaheuristic ensemble framework. Individual models are trained with task-specific loss functions (Asymmetric Loss, ZLPR) and per-label threshold tuning on the validation set. The ensemble fuses outputs through ten base strategies and four composition operators, with weights searched via random restarts, hill-climbing, and Variable Neighbourhood Search (VNS). Our best submission achieves a micro-F1 of **0.8667** on the official test set, improving by approximately 1.8 F1 points over the best standalone model.

Keywords: ICD-10 coding · clinical NLP · multi-label classification · BERT · ensemble methods · Greek language

1 Introduction

Clinical coding assigns standardised diagnostic codes to free-text medical records, a mandatory step in hospital administration, epidemiological surveillance, and insurance reimbursement. Manual coding is slow, costly, and prone to inter-annotator disagreement, motivating the development of automatic approaches [5].

The **ElCardioCC 2026** shared task targets this problem for Greek cardiology discharge summaries. Systems must predict all applicable ICD-10 codes for each document, a multi-label classification problem over 115 codes.

Three challenges define the setting. First, Greek is a morphologically rich, low-resource language for clinical NLP, with discharge summaries mixing Greek terminology and transliterated abbreviations (PCI, TAVI, CABG). Second, some summaries exceed the 512-token BERT limit. Third, the 115 ICD-10 codes span frequencies from a single instance to over 1,000.

No single architecture dominates across all labels and document types: neural models excel on frequent labels, dictionary rules on procedural codes, retrieval on rare coverage, and NER+EL on mention-level disambiguation. We build a system combining all four paradigms under a **declarative metaheuristic ensemble** that searches over 10 base strategies and 4 composition operators on the validation set.

2 Background

Automatic ICD coding from clinical text has been addressed with CNN/RNN architectures [12], BERT-based models [4,7], and hybrid retrieval-classification pipelines [5]. For Greek clinical NLP, Greek BERT [6] provides a domain-relevant starting point, while XLM-RoBERTa [1] handles Greek–Latin code-switching in clinical abbreviations. Label imbalance is the central challenge in multi-label coding: Asymmetric Loss (ASL) [11] addresses this by suppressing easy negatives more aggressively than positives. Ensemble methods [3] and VNS-based search [9] over non-differentiable objectives complement neural approaches for rare-label coverage.

3 Task and Dataset

3.1 Task Definition

ElCardioCC 2026 is a multi-label ICD-10 coding shared task for Greek cardiology discharge summaries (BioASQ/CLEF 2026), with 115 target codes. The primary metric is **relaxed micro-F1**: predictions from the same clinical cluster as the gold label are not fully penalised, reflecting clinical equivalence among related codes.

3.2 Dataset

The dataset consists of 2,500 anonymised discharge summaries split into training (2,000), validation (250), and test (250) sets, with a separate blind test set held by the organisers. Documents are moderately long: mean 280 tokens ($\approx$2,000 characters), maximum 1,218 tokens. Only 2.4% of documents exceed the 512-token BERT limit.

Label distribution. The 115 codes are highly skewed. The five most frequent (Z95, I25, I21, I48, E11) account for $\sim$41% of all annotations. At the other extreme, 30 codes appear fewer than 10 times and 4 codes appear only once in the entire dataset (Table 1). This long-tailed distribution severely challenges standard classifiers and requires explicit strategies for rare-label coverage.

3.3 Preprocessing and Data Augmentation

Text cleaning. Shared across all splits: whitespace collapse, removal of non-essential special characters (preserving hyphens, commas, slashes, parentheses). Greek BERT additionally receives lowercasing and accent stripping; XLM-RoBERTa

Table 1. Label frequency distribution across 115 ICD-10 codes.

Frequency band	# codes	Approx. % of instances
≥ 500	5	41%
100–499	22	36%
10–99	58	21%
< 10	30	<2%

receives whitespace-only normalisation to preserve abbreviation signals ("PCI", "TAVI"). ICD-10 annotations are encoded as 115-dimensional binary multilabel vectors.

Stratified splitting. A two-stage Multilabel Stratified Shuffle Split [13] produces an 80/10/10 partition, ensuring every label present in validation or test also appears in training. Of the 2,500 samples, 1,453 include mention-level annotations enabling NER+EL training.

Synonym augmentation (XLM-RoBERTa base only). Dictionary terms are swapped with clinical synonyms at $p = 0.35$ using frequency-adaptive multipliers (×4 for 30–99 instances, ×3 for 100–199, ×2 for 200–299).

4 Methodology

The system pipeline is: (i) train six independent components; (ii) tune per-model decision thresholds on the validation set; (iii) search for the best ensemble fusion strategy. Figure 1 provides an overview.

Raw JSONL (train / validation / test)

↓

Preprocessing & Stratified Split

↓

Greek BERT	XLM-R Large	XLM-R Base	Hybrid IR	Dictionary Baseline	NER +EL

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Per-model threshold tuning (validation set)

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Metaheuristic Ensemble (10 strategies + 4 compositions)

↓

Final Predictions

Fig. 1. High-level system pipeline.

4.1 Greek BERT Multi-Label Classifier

Architecture. We fine-tune `nlpaueb/bert-base-greek-uncased-v1` (Greek BERT) [6] with a two-layer MLP head: mean pooling $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(768$\rightarrow$384, GELU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(384$\rightarrow$115). Mean pooling outperformed CLS-token extraction by $\sim$0.8 F1 points [12], as diagnostic evidence is distributed across the full document. Max sequence length: 384 tokens.

Training. Table 2 lists the final hyperparameters. **ASL** [11] replaced BCE after Phase 2, raising validation micro-F1 from $\approx$0.74 to 0.788 by suppressing easy negatives. ASL applies asymmetric focusing to positive and negative terms:

$$\mathcal{L}_{\text{ASL}} = \sum_{y} \begin{cases} (1-p)^{\gamma^+} \log p & y = 1 \\ (p_m)^{\gamma^-} \log(1-p_m) & y = 0 \end{cases} \tag{1}$$

where $p_m = \max(p - m, 0)$ shifts and clips easy negatives, $\gamma^+ = 1$, $\gamma^- = 5$, $m = 0.05$. **LLRD** (factor 0.85) updates lower encoder layers at $\text{LR} \times 0.85^l$, preserving pre-trained Greek representations while adapting upper layers to the clinical domain; gain $\approx$+0.4 F1. The model evolved across four phases (41 WandB runs): BCE baseline (0.74) $\rightarrow$ ASL (0.788) $\rightarrow$ LLRD + MLP (0.811) $\rightarrow$ aggressive tuning (0.827 base, **0.854** after per-class threshold tuning).

Table 2. Greek BERT final hyperparameters.

Hyperparameter	Value
Max length / batch	384 tok / 32 (16 + grad. accum. 2)
Learning rate / LLRD	4×10^{-5} / 0.85 per layer
Warmup / weight decay	6% / 0.02
Loss	ASL ($\gamma^- = 5$, $\gamma^+ = 1$, clip = 0.05)
Epochs / patience	30 / 7 (FP16)

Threshold tuning. A two-pass sweep selects $t_l^* = \arg\max_{t \in [0.45, 0.75]} F1_l(t; \mathcal{D}_{\text{val}})$ per label independently. Pass 2 accepts per-label values only for labels with $\geq$15 positive examples and ΔF1 > 0.001. This adds $\approx$2–3% F1, bringing the tuned validation micro-F1 to **0.854** (ensemble context: **0.8165**). Figure 2 (after all components) shows the full phase progression.

4.2 XLM-RoBERTa Classifiers

XLM-RoBERTa's 250K multilingual vocabulary handles Greek–Latin code-switching better than Greek BERT's 35K vocabulary.

Long-document strategy. A sliding window (stride 256) handles documents exceeding 512 tokens. At inference, chunk logits are fused as $\hat{y} = 0.5\,\bar{y}_{\text{mean}} + 0.5\,y_{\text{max}}$, balancing average evidence with the most confident chunk signal.

Large variant. ZLPR loss, effective batch 32 (batch 2, grad. accum. $\times 16$), LR 2×10^{-5}, weight decay 0.05, dropout 0.15, 40 epochs (patience 5). Validation F1 ceiling $\approx$0.76; slower convergence and memory constraints (560M parameters) led us to focus development on Greek BERT.

Base variant. Implements *MedicalModelWithDescriptions*: the `[CLS]` representation passes through 5 parallel dropout layers ($p = 0.1i$, $i \in \{1 \ldots 5\}$) whose logits are averaged (**multi-sample dropout**). A **semantic anchoring** term adds cosine similarity between the document embedding and frozen ICD-10 description embeddings, scaled by a learnable scalar α:

$$\hat{y} = W\mathbf{h} + \alpha \cdot \tfrac{\mathbf{h}}{\|\mathbf{h}\|}\mathbf{D}^{\top} \tag{2}$$

α converges from 0.099 to $\approx$0.12, confirming a consistent but small semantic correction. Training: AdamW LR 2×10^{-5}, early stopping (patience 4), per-class positive weights (ceiling 20). Post-processing: dynamic threshold tuning + ICD-10 hierarchy enforcement + co-occurrence rules. Validation micro-F1: **0.8101**. Despite using a smaller backbone than the Large variant, the Base model outperforms it (0.810 vs. 0.748) due to the semantic anchoring mechanism and task-specific post-processing, while the Large variant was constrained to batch size 2 under GPU memory pressure.

4.3 Information Retrieval Module

Three retrieval channels (BM25 [10], TF-IDF, dense MiniLM embeddings) are fused by Reciprocal Rank Fusion [2] ($\text{RRF}(d) = \sum_i 1/(k + r_i(d))$, $k = 60$). Codes with RRF score $\geq 0.22 \times$ top score, capped at 12 per document, are predicted. The cap is critical: raising it to 25 collapses F1 from 0.662 to 0.140 due to precision loss.

4.4 Dictionary Baseline

A curated CSV (1,680 term-to-code entries) is matched by an Aho–Corasick automaton with a 35-term blacklist. Cardiology procedure rules fire on keywords such as PCI/PPCI/TAVI/CABG (adding Y84, Z95, and related codes); co-occurrence rules enforce clinical consistency (e.g. I22 $\Rightarrow$ I21, I25 $\Rightarrow$ Z95). Validation micro-F1 **0.7176** (precision 0.65, recall 0.80), covering 99.76% of documents.

4.5 Named Entity Recognition and Entity Linking

Greek BERT is fine-tuned as a BIO sequence tagger with a single entity type (`MED`) and a Partial CRF [14]. The coarse type maximises recall; the Partial CRF enforces label consistency via Viterbi decoding and enables semi-supervised learning from documents with only document-level annotations. Training: 6 epochs, batch 8, LR 2×10^{-5}, early stopping patience 2.

NER spans are augmented by Aho–Corasick dictionary matching; containment-based deduplication retains the longest span. Each mention m is linked to ICD-10 code c by:

$$\text{score}(m, c) = 0.6 \cdot P_{\text{prior}}(c \mid m) + 0.4 \cdot \cos(\mathbf{e}_{\text{ctx}}, \mathbf{e}_c) \tag{3}$$

where P_{prior} is the (mention, code) co-occurrence frequency from training and $\mathbf{e}_{\text{ctx}}$, $\mathbf{e}_c$ are MiniLM embeddings of a ±200-char context window and the Greek ICD-10 description respectively. Full document-level dictionary boosting is applied after mention linking. Best validation micro-F1: **0.7223** at epoch 5.

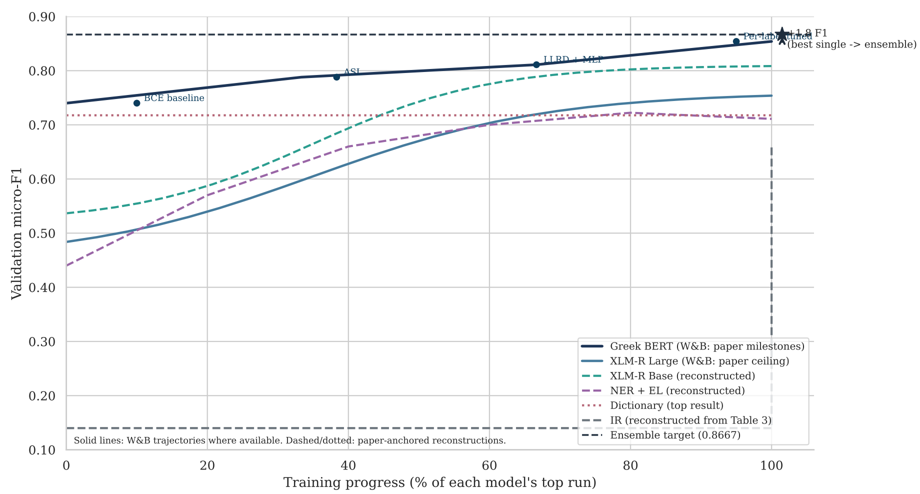

Fig. 2. Validation micro-F1 progression for all six components across training. Greek BERT milestones are annotated (BCE baseline → ASL → LLRD+MLP → final tuned); the dashed line marks the ensemble target (0.8667).

4.6 Metaheuristic Ensemble

Rather than fixing a single fusion rule [3], we define a **declarative strategy space** in YAML and search over it on the validation set, enabling the addition of new strategies without code changes.

Base strategies. Nine base strategies cover the main fusion paradigms. The primary strategy is the *weighted ensemble*, a convex combination of all component probability vectors:

$$\hat{y} = \sum_{i=1}^{6} w_i \, \hat{y}^{(i)}, \quad w \in \mathcal{W} = \Big\{ w \in \mathbb{R}^6 : \textstyle\sum_i w_i = 1, \; w_i \geq 0 \Big\} \tag{4}$$

The remaining strategies address complementary failure modes: *per-label champion* routes each label to its highest-validation-F1 component; *gated secondary* falls back to IR/NER for documents where the neural model is uncertain; *frequency-bucketed thresholding* applies separate decision thresholds per label-frequency band; and *label correction* mines co-occurrence rules ($a \to b$) from validation predictions, predicting label b whenever label a is predicted with confidence above a mined threshold.

Composition operators. Rather than selecting a single base strategy, three label-set algebra operators compose multiple strategy outputs $\{\hat{L}^{(s)}\}_{s \in \mathcal{S}}$, where $\hat{L}^{(s)} = \{l : \hat{y}_l^{(s)} \geq t_l\}$:

$$\hat{L}^{\mathrm{or}} = \bigcup_{s \in \mathcal{S}} \hat{L}^{(s)} \qquad \text{(maximises recall)} \tag{5}$$

$$\hat{L}^{\mathrm{and}} = \bigcap_{s \in \mathcal{S}} \hat{L}^{(s)} \qquad \text{(maximises precision)} \tag{6}$$

$$\hat{L}^{k\text{-of-}n} = \Big\{ l : \sum_{s \in \mathcal{S}} \mathbb{1}[l \in \hat{L}^{(s)}] \geq k \Big\} \qquad \text{(balances precision/recall)} \tag{7}$$

The best submission applies *merge_ and* over the weighted ensemble and the label correction module, letting the correction step surgically remove false positives from the ensemble output.

Weight optimisation. Weights are optimised directly on the non-differentiable validation micro-F1 objective:

$$w^* = \arg\max_{w \in \mathcal{W}} \mathrm{F1}_\mu\Big(\hat{y}(w), \, y^{(\mathrm{val})}\Big) \tag{8}$$

in two stages: (1) 10,000 random vectors from $\mathcal{W}$, each refined by pairwise hill-climbing; (2) VNS [9] from the best solution, widening the neighbourhood radius when no improvement is found and resetting when one is.

Submitted strategies. The five submissions span the precision–recall trade-off: standalone Greek BERT (80/10 split and 100% data) establish the single-model ceiling; three ensemble submissions apply *merge_ and* (precision-oriented), *merge_ k2* (balanced), and a safer-threshold *merge_ k2* variant. The *merge_ and* composition achieves the best F1 by combining the weighted ensemble's strong baseline with the label correction module's precision boost.

5 Experimental Results

5.1 Individual Component Performance

Table 3 reports standalone validation micro-F1 for each component. Greek BERT is the strongest single model by a large margin. Dictionary and NER+EL perform comparably; the IR module has the lowest F1 but the highest recall (0.83), making it a key coverage contributor in the ensemble.

Table 3. Individual component validation micro-F1.

Component	Val. micro-F1	Notes
Greek BERT (per-label thresholds)	0.8165	Best single model
XLM-RoBERTa Base (tuned)	0.8101	Semantic label anchoring
XLM-RoBERTa Large	≈0.76	ZLPR loss; slower convergence
NER + Entity Linking	0.7223	Best at epoch 5
Dictionary Baseline	0.7176	99.76% document coverage
IR (hybrid RRF)	0.6621	Recall 0.69, Precision 0.63

5.2 Ablation Studies

Threshold tuning. Table 4 shows the effect of progressively more granular thresholding. Per-label tuning adds ≈1.5 F1 points over a fixed threshold of 0.5 and ≈1.0 points over the global optimum.

Table 4. Threshold tuning ablation for Greek BERT.

Strategy	Val. micro-F1
Fixed $t = 0.5$	≈0.79
Global sweep $[0.45, 0.75]$	≈0.80
Per-label sweep (2-pass)	**0.8165**

Impact of LLRD. Layer-wise learning rate decay improved validation F1 by ≈0.4 points. The improvement is most pronounced for rare labels, where preserving pre-trained lower-layer representations is more important than task-specific adaptation.

Loss function comparison. ASL outperformed both binary cross-entropy and standard focal loss, primarily through better recall on rare labels. BCE delivered good precision but lower recall; focal loss improved recall relative to BCE but less than ASL due to its symmetric treatment of positive and negative examples.

Ensemble composition. Table 5 compares the three composition operators at a schematic level. The intersection (`merge_and`) achieves the best F1 by combining the weighted ensemble's strong baseline with the correction module's precision improvement.

Table 5. Ensemble composition operator comparison (validation set, schematic).

Strategy	Recall	Precision	F1 trend
Union (merge_or)	highest	lowest	moderate
Weighted ensemble	mid	mid	strong
k-of-n (merge_k2)	mid	mid+	strong
Intersection (merge_and)	mid−	highest	**best**

5.3 Official Test Set Results

Table 6 reports all five submissions on the official test set. The best submission uses the `merge_and` composed strategy (weighted ensemble intersected with the label correction module).

Table 6. Official ElCardioCC 2026 test set results.

Submission	Recall	Precision	F1
ensemble (merge_and: weighted + correction)	**0.8510**	**0.8830**	**0.8667**
ensemble (merge_k2: weighted + per-label + corr.)	0.8687	0.8539	0.8613
ensemble (safer thr, merge_k2)	0.8767	0.8431	0.8596
mlc_greek_bert_100 (100% data, no val. split)	0.8194	0.8811	0.8491
mlc_greek_bert (80/10/10 split)	0.8310	0.8675	0.8489

The ensemble outperforms the best standalone model by **+1.8 F1 points** (0.8667 vs. 0.8489), confirming the complementarity of the six components. The full-data Greek BERT (trained on 100% of available data without a validation split) achieves virtually the same F1 as the standard split version (0.8491 vs. 0.8489), suggesting that the validation set contributes little additional signal at this scale.

Among ensemble submissions, the precision-oriented `merge_and` yields the highest overall F1 (0.8667), while `merge_k2` variants achieve higher recall at the cost of precision, consistent with the 2-of-3 voting design. The safer threshold variant of `merge_k2` reaches the highest recall (0.8767) but the lowest precision (0.8431).

5.4 Error Analysis

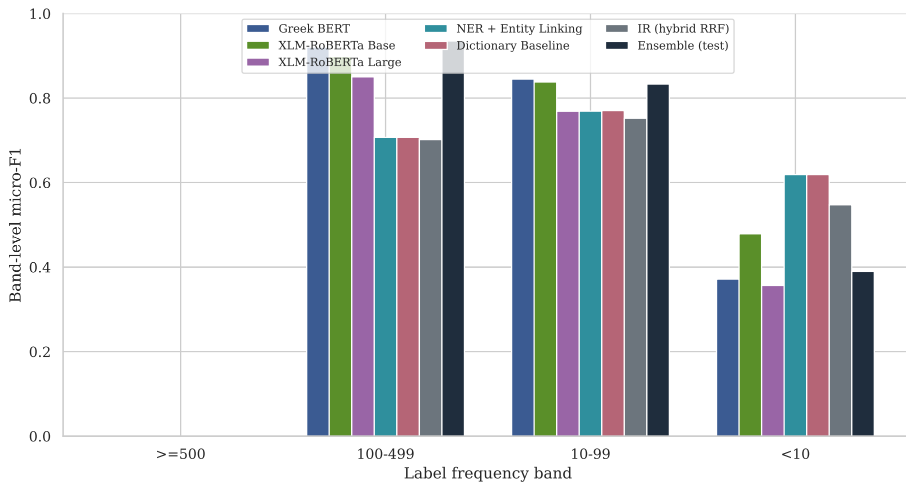

Fig. 3. Band-level micro-F1 per component across label frequency groups. All models degrade sharply on rare labels (<10 instances); the ensemble provides no recovery for this band.

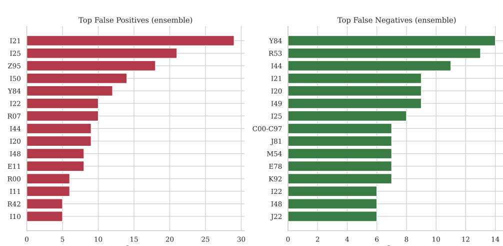

Fig. 4. Top false-positive and false-negative codes for the best ensemble submission.

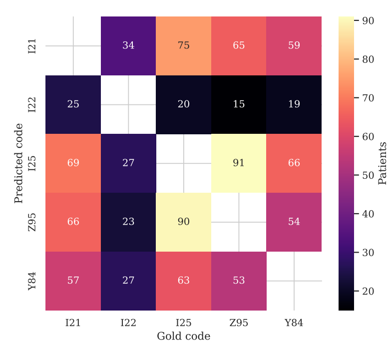

Fig. 5. Co-prediction counts for the MI/procedure cluster (I21, I22, I25, Z95, Y84).

Rare labels. The 30 codes with <10 training instances are the primary error source. For 10 of these, per-label thresholding is not applicable. The dictionary rule-based system recovers some (e.g. E85 via the amyloidosis rule), but codes with no lexical surface form (e.g. M35, R57) remain consistently unrecovered.

Coverage gaps. 31 label types lie outside the dictionary's coverage, including metabolic (E00), haematological (D47), and musculoskeletal codes. These require inference from laboratory values or implicit multi-sentence evidence.

Label confusion. I21/I22/I25 (acute MI, subsequent MI, chronic ischaemic disease) frequently co-occur in a single document and require temporal reasoning to assign correctly, which is beyond the capability of current bag-of-token models. Z95 (presence of vascular implants) is the main false-positive source due to ubiquitous stent and pacemaker mentions, motivating the `merge_and` correction step.

Calibration. Greek BERT tends to under-predict rare labels: true positive probabilities for rare codes often fall below 0.5. Per-label thresholding helps but becomes unreliable for labels with fewer than 15 positive validation examples.

6 Conclusion

We presented a multi-component system for automatic ICD-10 coding of Greek cardiology discharge summaries, developed as part of the MSc in Artificial Intelligence and Data Analytics at the University of Macedonia.

The system integrates Greek BERT and XLM-RoBERTa neural classifiers with a dictionary-based rule system, a hybrid information retrieval module, and a NER+EL pipeline. These are fused by a declarative metaheuristic ensemble that searches over ten base strategies and four composition operators using random search, hill-climbing, and Variable Neighbourhood Search. Our best submission achieves micro-F1 **0.8667**, a gain of 1.8 points over the standalone Greek BERT model.

The error analysis identifies three main limitations: rare-label under-coverage due to insufficient training data and calibration issues; dictionary gaps for non-cardiology comorbidity codes; and label confusion among temporally related ICD-10 codes.

Future work will target rare-label calibration (temperature scaling, isotonic regression), data augmentation for low-frequency codes, and temporal reasoning modules for distinguishing related codes such as I21/I22/I25.

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